

Human Behaviour in Evacuation Dynamics

Evacuation scenarios are characterized by complex, adaptive human behaviours that deviate from simple shortest-path motion. Under stress or time pressure, individuals may panic, push, clog exits or even slow down crowd flow ("faster-is-slower" effect) ¹. For example, Helbing et al. showed that when everyone tries to move faster, competition causes arching and clogging at exits, actually slowing the evacuation ¹. Similarly, in real fires people often change from calm to aggressive behaviour (e.g. running at a red light) and may stampede or freeze as stressors change ² ¹. These studies highlight that panic and stress can lead to erratic, emergent crowd dynamics (see image below). Moreover, simulation reviews note that **most optimization schemes still assume idealized agents**, focusing on shortest routes, whereas agent-based models show rich heterogeneity ³. To avoid overly optimistic GA solutions, one must incorporate human factors such as variable mobility, social influence, and decision biases into the GA.

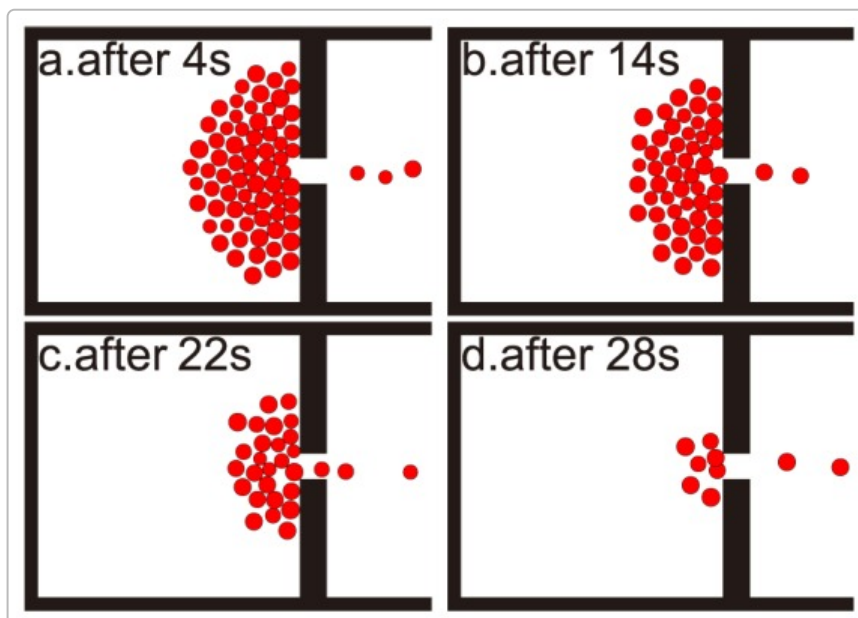


Figure: Simulated panic evacuation in a crowded hall (from Jiang et al.), illustrating congestion and pushing at the exit. In panic conditions agents tend to "faster-is-slower" effects and complex flows ¹ ².

Key Human Traits in Evacuation

Human evacuees differ in many traits that affect route choice and timing. Key factors found in the literature include:

- **Heterogeneous mobility and vulnerability:** Individuals vary widely in speed and ability. Elderly, injured or disabled agents (e.g. wheelchair users, visually impaired) move more slowly or cannot use all routes ⁴ ⁵. For example, one study found that adding 20% disabled agents (wheelchairs/blind) increased total evacuation time by ~50% and cut peak throughput ~40% ⁴. These mobility constraints (e.g. cannot climb stairs) must be respected in the solution encoding.
- **Social grouping and leadership:** People often evacuate in groups (families or friends). Close-knit groups tend to stick together: parents will not leave without children, friends may wait for

each other ⁶. Sime (1983) observed that individuals with strong social ties were less likely to panic and chose group-optimal actions (even at personal risk) ⁶. Group behavior can include collectively choosing a leader to follow ⁷.

- **Altruism and helping:** Many agents will assist others if possible. Models of altruistic evacuation assign a probability that nearby healthy agents will stop and help fallen or disabled evacuees ⁸. Helping slows the helper's progress (reducing speed by a factor) and delays both agents' evacuation ⁸. Studies show these helping dynamics significantly affect evacuation efficiency (delaying exits) depending on "help probability" and range ⁹.
- **Panic and emotional contagion:** Anxiety can spread through a crowd. Panic tends to make mutation in decision-making (e.g. people abandon a plan and try something new, or move randomly). Empirical reports suggest true panic is rare, but stressors (smoke, alarms, crowding) do induce changes: people may follow leads or, if no one is around, wander or freeze ¹ ⁶. Individuals with weak social ties or low group affinity panic more easily ⁶.
- **Knowledge and familiarity:** Some evacuees (staff, regular commuters) know the building layout or exits well, while others are disoriented. Familiarity affects path choice: experienced agents may head straight to known exits, whereas others may search or follow crowd cues ⁵. Cognitive biases (normalcy bias, optimism) can delay reactions ¹⁰.

Mapping Traits to GA Operator Modifications

We can incorporate the above traits by **modifying GA operators** so that agent "chromosomes" behave more realistically. Possible approaches include:

- **Heterogeneous mobility constraints:** Encode each agent's capabilities into the chromosome and operators. For example, for a wheelchair agent, disallow mutation or crossover that inserts paths with stairs. In mutation, replace a forbidden segment with an accessible detour. During fitness evaluation, use agent-specific speeds (slower for disabled) ⁴. This ensures the GA does not produce unrealistically fast paths for low-mobility agents.
- **Group affiliation / leadership:** Modify crossover/selection to preserve group coherence. If two agents belong to a group, one could designate a "leader" chromosome gene whose route is copied or given priority. For instance, implement a *group-based crossover*: when crossing two solutions, swap entire subsets of agent routes for group members rather than mixing them. Or use *crossover elitism*: keep one agent's path (the leader) fixed and only mutate followers. This mimics how in reality members tend to copy the leader's decision ⁷ ⁶.
- **Altruism (cooperative behavior):** Introduce cooperative genetic operators. For example, allow *biased crossover* where a helper agent's beneficial genes (fast exit segments) are shared with a vulnerable agent. Or use a *memetic adjustment*: if an agent is marked "vulnerable", apply a local search operator that ensures a helper's path defers (adds waiting or rerouting to accommodate). Another idea is multi-objective selection: optimize for both overall time and fairness, so solutions that delay a strong agent to help a weak one are considered fitter in altruistic contexts ⁸.
- **Panic / stress adaptation:** Build *adaptive mutation strategies*. Already panic can increase an agent's mutation rate (randomness) ⁹. Similarly, one could make **mutation and crossover rates dynamic**: when simulated stress is high (e.g. many agents crowded), increase mutation globally to simulate erratic decisions. Alternatively, use a *catastrophe operator*: if panic threshold exceeded, partially or fully re-randomize a chromosome's paths to mimic abandoning a plan. High "panic" agents may also preferentially choose local exit segments (greedy escape moves).
- **Herding and social influence:** Emulate social learning by sharing good solutions. For example, implement a *cultural or population-level operator*: periodically replace poorer chromosomes with copies of segments from the current best solution, as if less-informed agents "follow the crowd." Another approach is **co-evolutionary crossover**: allow agents from different chromosomes to

swap routes if one route is generally more popular or faster, simulating a “herding” effect. This mimics how individuals might follow routes chosen by the majority.

- **Familiarity / memory:** Give certain agents a “knowledge” advantage via local optimization. For example, apply a path-shortening local search (like Dijkstra on a subgraph) selectively for agents marked as familiar. Conversely, less-familiar agents could have higher exploration (mutation) rates. You could also “seed” initial population with the known-good route for one agent (if familiar) and let GA refine the rest. This mimics how trained evacuees quickly pick an optimal route while others meander.

Each trait-driven modification alters the GA’s behavior. For instance, **introducing group-based crossover** tends to keep families’ routes similar, reducing unrealistic splitting of groups, while **adaptive mutation for panic** injects variability to avoid overconfident timing. In practice, one would test these operator variants on benchmark evacuation graphs and compare outputs. We expect that adding human-like traits (as supported by simulation studies) will **slow down evacuation times and add realism** relative to a naive GA ³ ⁴. For example, accounting for disabled agents alone was shown to lengthen evacuation by ~50% ⁴, so a GA that includes such traits will predict longer, more varied escape times than an optimistic all-fast model.

Sources: Reviewed evacuation studies highlight factors like panic dynamics ¹ ², group cohesion ⁶ ⁷, altruistic aid ⁸ and mobility heterogeneity ⁴ ⁵. These inform the design of GA operators above. A recent survey notes that most routing optimizers still ignore these behaviors, underscoring the need to integrate them ³.

¹ ⁴ ⁵ Inclusive crowd evacuation modeling under heterogeneous mobility constraints | Scientific Reports

https://www.nature.com/articles/s41598-025-19403-x?error=cookies_not_supported&code=fd955c20-9238-4564-bdea-8d2162699560

² Interactive Simulation of Dynamic Crowd Behaviors using eneral Adaptation Syndrome Theory

<https://gamma.cs.unc.edu/GAScrowd/Stressor.pdf>

³ A systematic review of methodologies for human behavior modelling and routing optimization in large-scale evacuation planning - ScienceDirect

<https://www.sciencedirect.com/science/article/pii/S221242092400400X>

⁶ ⁷ repository.gatech.edu

<https://repository.gatech.edu/bitstreams/da5b1342-d4ea-40e9-9ea5-df5fea37560f/download>

⁸ ⁹ systems.enpress-publisher.com

<https://systems.enpress-publisher.com/index.php/jipd/article/viewFile/8184/4046>

¹⁰ Guidance for the Model User on Representing Human Behavior in Egress Models - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5292781/>